Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3481-DE01

Title: (BEng/MEng)

Power Semiconductor Devices

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Q1

A piece of silicon is uniformly doped with donors. At equilibrium conditions, the Fermi level is 0.3 eV above the intrinsic Fermi level.

(a) i. Explain what is meant by 'equilibrium' conditions and how they differ from 'steady-state' conditions.

ii. What is the density of equilibrium electrons, n_0, and holes, p_0, in the silicon?

[12%]

(b) The silicon is now subjected to illumination that causes uniform carrier generation. The light intensity, and thus the generation rate, increases linearly with time from zero at $t = 0$ to the full generation rate G_L at $t = 2\tau_p$, where τ_p is the hole lifetime.

i. Write an expression for the density of excess of carriers as a function of time for the time interval between $t = 0$ and $t = 2\tau_p$, using G_L and τ_p as parameters. Hint: express your differential equation as $y'(x) = ax - by(x)$, for which the solution is:

$$y(x) = -\frac{a}{b^2} + \frac{a}{b}x + c_1 \exp(-bx)$$

Where c_1 is an integration constant.

ii. What is the density of excess of carriers at $t = 2\tau_p$? Express your answer as a percentage of the expected steady-state density.

[32%]

(c) i. From $t = 2\tau_p$ onwards the sample is under constant illumination producing a generation rate of G_L. Derive an expression for the density of excess of carriers in the period of time from $t = 2\tau_p$ onwards. Hint: arrange your differential equation in the form $y'(x) = a - by(x)$, for which the solution is:

$$y(x) = \frac{a}{b} + c_2 \exp(-bx)$$

Where c_2 is an integration constant. *If you did not reach a solution to Q1.b, you may use* $\Delta p(2\tau_p) = 0.6\Delta p(\infty)$

ii. What is the steady state density of charge carriers if $G_L = 6 \times 10^{18}$ cm^{-3}s^{-1} and $\tau_p = 0.5$ μs? Are the conditions of low injection still valid here?

[32%]

(d) When used as a photodetector, the "rise time" is defined as the time it takes for the excess of carriers to reach 95% of their steady state density, *i.e.*, $\Delta p_{0.95} = 0.95\Delta p_\infty$. Calculate the rise time based on the equations found in parts (b) and (c) **[24%]**

Q2

A thin chip of gallium arsenide (GaAs) is doped with donors at a density of $N_D = 8 \times 10^{15}$ cm^{-3}. A layer of chromium (Cr, $\phi_M = 4.5$ eV) was evaporated uniformly on the top side of the chip, and on top of that, a single crystal thin film of GaAs was grown and doped with acceptors at a density of $N_A = 2 \times 10^{17}$ cm^{-3}. The dimensions of the structure are shown in **Fig. Q2**.

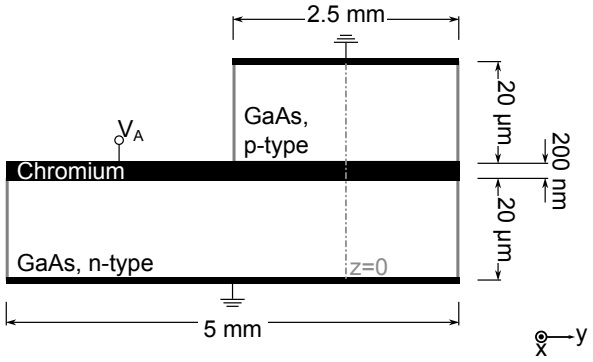


Fig. Q2

The bottom side of the chip and the top side of the GaAs film are grounded through perfectly Ohmic contacts, and the chromium plate is biased with V_A. The lengths of the chip, electrodes and top GaAs layer in the direction normal to the plane of the figure is 2.5 mm.

(a) i. For each Cr / GaAs interface, determine if it is a Schottky or an Ohmic contact.
ii. For any Schottky interface, determine the built-in potential. Give your answer in (V).
iii. Draw a schematic band diagram of the device along the dashed line in **Fig. Q2**.

[20%]

(b) What is the total capacitance that will be measured between V_A and the ground, when no DC bias is applied to the chromium ($V_A = 0$ V)? Give your answer in nF. Neglect the capacitance of any Ohmic contact. *If you did reach a solution to Q2.a, you may use* $V_{bi}^{(p)} = 0.95$ *V,* $V_{bi}^{(n)} = 0.40$ *V.* **[36%]**

(c) i. What is the total capacitance that will be measured between V_A and the ground, when DC bias values of $+0.25$ V and -0.25 V are applied with respect to the ground?
ii. Explain the difference between the C-V trend you observe to the one seen in conventional pn junctions.

[24%]

(d) We now produce an identical sample, but with platinum (Pt, $\phi_M = 5.5$ eV) instead of chromium.

i. Draw a schematic band diagram of the new device along the dashed line in **Fig. Q2**.
ii. Calculate the current through the device when $V_A = 0.1$ V, if the modified Richardson constants, $\mathscr{A}^*$, for a GaAs/Pt contact are 0.88 and 1.37 A cm^{-2}K^{-2} for p-type and n-type, respectively.

[20%]

Q3

A silicon carbide p-i-n diode is designed to support the operation of maintenance robots in fusion energy and has a maximum operating temperature of 1000 $^{\circ}$C.

(a) Determine the minimum dopant concentration in the drift region and subsequently calculate the specific on–state resistance of the diode if the thickness of the drift region is 25 μm. **[20%]**

(b) For the diode in part a, estimate the change in the minority carrier concentration in the n–type region when the diode is operated at half of the built–in potential, i.e. $\frac{V_{BI}}{2}$. Use this to predict the injected charge density in to the diode during a switching transient. **[25%]**

(c) For the diode in part a, determine the non–punchthrough blocking voltage and the maximum electric field in the diode when operating in blocking. **[20%]**

(d) The diode is part of a motor control circuit where the switching frequency is 150 kHz. The operator of the fusion reactor has noticed that the diodes are failing because of excessive junction temperature, which is related to the power losses. What advice would you offer to the operator in terms of the choice of diode for future upgrades? **[35%]**

Q4

A schematic cross–section of a silicon carbide lateral Junction Field Effect Transistor (JFET) is shown in **Fig. Q4**. The channel region of the JFET is manufactured from silicon carbide with a dopant concentration $N_D = 1 \times 10^{17}$ cm^{-3}, thickness of the n$^-$ drift region of 1 μm and a gate length, L_G of 5 μm. The back gate (which is at the lower edge of the figure) is electrically connected to the source contact

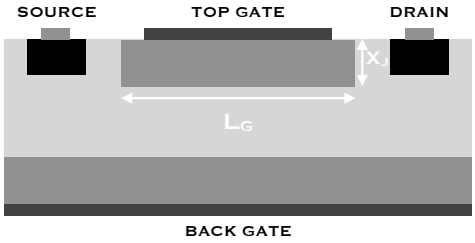


Fig. Q4

(a) Outline a suitable fabrication process for the JFET structure shown in **Fig. Q4** **[25%]**

(b) Determine the depth of the gate implant, denoted by X_J in **Fig. Q4**, if the threshold voltage of the JFET is –5 V. **[20%]**

(c) Calculate the On–state resistance of the JFET when $V_{GS} = 2$ V and the gate width is 2 mm. **[25%]**

(d) The initial testing of the manufactured JFET shows a variation in the dopant concentration and the drift region thickness of $\pm$ 10 %. Determine the resulting shift in the threshold voltage and hence on–state resistance of the JFET when $V_{GS} = 0$ V. **[30%]**

ENGI3481-WE01 Power Semiconductor Devices
Formula sheet - Questions 1 & 2

$$v_g = \frac{\omega}{k} \tag{1}$$

$$\omega = 2\sqrt{\frac{\beta}{m}} \cdot \left|\sin\left(\frac{1}{2}ka\right)\right| \tag{2}$$

$$\omega = |k|\, a\sqrt{\frac{\beta}{m}} \tag{3}$$

$$\lambda_{\max} = \frac{hc}{E_g} \tag{4}$$

$$\lambda\nu = c \tag{5}$$

$$f_{FD}(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{k_BT}\right)} \tag{6}$$

$$m^* = \pm\hbar^2\left(\frac{\partial^2 E}{\partial k^2}\right)^{-1} \tag{7}$$

$$n = N_C \exp\left(-\frac{E_C - E_F}{k_BT}\right) \tag{8}$$

$$n = n_i \exp\left(\frac{E_F - E_i}{k_BT}\right) \tag{9}$$

$$p = N_V \exp\left(-\frac{E_F - E_V}{k_BT}\right) \tag{10}$$

$$p = n_i \exp\left(\frac{E_i - E_F}{k_BT}\right) \tag{11}$$

$$N_C = \frac{1}{4}\left(\frac{2m_n^* k_BT}{\pi\hbar^2}\right)^{\frac{3}{2}} \tag{12}$$

$$N_C = 2.5 \cdot \left(\frac{m_n^*}{m_e}\right)^{\frac{3}{2}} \cdot \left(\frac{T}{300K}\right)^{\frac{3}{2}} \times 10^{19} cm^{-3} \tag{13}$$

$$N_V = \frac{1}{4}\left(\frac{2m_p^* k_BT}{\pi\hbar^2}\right)^{\frac{3}{2}} \tag{14}$$

$$N_V = 2.5 \cdot \left(\frac{m_p^*}{m_e}\right)^{\frac{3}{2}} \cdot \left(\frac{T}{300K}\right)^{\frac{3}{2}} \times 10^{19} cm^{-3} \tag{15}$$

$$n \cdot p = n_i^2 \tag{16}$$

$$n_i = \sqrt{N_C N_V}\exp\left(-\frac{E_g}{2k_BT}\right) \tag{17}$$

$$E_i = \frac{E_C + E_V}{2} - \frac{3}{4}k_BT\ln\left(\frac{m_n^*}{m_p^*}\right) \tag{18}$$

$$J = \sigma\mathscr{E} \tag{19}$$

$$\sigma = q\left(\mu_n n + \mu_p p\right) \tag{20}$$

$$\mu = \frac{q\tau}{m^*} \tag{21}$$

$$J_n = q\mu_n n\mathscr{E} + qD_n\frac{dn}{dx} \tag{22}$$

$$J_p = q\mu_p p\mathscr{E} - qD_p\frac{dp}{dx} \tag{23}$$

$$\frac{D}{\mu} = \frac{k_BT}{q} \tag{24}$$

$$\frac{dn}{dt} = \frac{1}{q}\frac{dJ_n}{dx} - R_n + G_n \tag{25}$$

$$N_A^- + n = N_D^+ + p \tag{26}$$

$$E_F - E_i = \tfrac{1}{2}k_BT \cdot \ln\left(\frac{n}{p}\right) \tag{27}$$

$$V_{bi} = \frac{k_BT}{q} \cdot \ln\left(\frac{N_A N_D}{n_i^2}\right) \tag{28}$$

$$W = \left[\frac{2\varepsilon_r\varepsilon_0}{q}\left(\frac{N_A + N_D}{N_D N_A}\right)(V_{bi} - V_A)\right]^{\frac{1}{2}} \tag{29}$$

$$I = I_S\left(\exp\left(\frac{qV_A}{k_BT}\right) - 1\right) \tag{30}$$

$$I_S = qA\left(\frac{D_n}{L_n}\frac{n_i^2}{N_A} + \frac{D_p}{L_p}\frac{n_i^2}{N_D}\right) \tag{31}$$

$$I = I_S\left(\exp\left(\frac{qV_A - IR_s}{k_BT}\right) - 1\right) \tag{32}$$

$$C = A\left[\frac{q\varepsilon_r\varepsilon_0}{2\left(V_{bi} - V_A\right)}\frac{N_D N_A}{N_D + N_A}\right]^{\frac{1}{2}} \tag{33}$$

$$Z = \frac{1}{j\omega C} \tag{34}$$

continued

$$R_d = \frac{dV}{dI} = \frac{k_B T}{q\left(I + I_S\right)} \tag{35}$$

$$V_{bi} = \frac{\Phi_{Bn}}{q} + \frac{k_B T}{q} \cdot \ln\left(\frac{N_D}{N_C}\right) \tag{36}$$

$$V_{bi} = \frac{\Phi_{Bp}}{q} + \frac{k_B T}{q} \cdot \ln\left(\frac{N_A}{N_V}\right) \tag{37}$$

$$\Phi_{Bn} = \Phi_M - \chi \tag{38}$$

$$\Phi_{Bp} = \chi + E_g - \Phi_M \tag{39}$$

$$W = \sqrt{\frac{2\varepsilon_r \varepsilon_0}{qN_D}\left(V_{bi} - V_A\right)} \tag{40}$$

$$C = A\sqrt{\frac{q\varepsilon_r \varepsilon_0 N_D}{2\left(V_{bi} - V_A\right)}} \tag{41}$$

$$I = I_S\left(\exp\left(\frac{qV_A}{k_B T}\right) - 1\right) \tag{42}$$

$$I_S = A\mathscr{A}^* T^2 \cdot \exp\left(-\frac{\Phi_B}{k_B T}\right) \tag{43}$$

	n-type	**p-type**
$\Phi_M > \Phi_S$	Rect.	Ohm.
$\Phi_M < \Phi_S$	Ohm.	Rect.

continued

ENGI3481-WE01 Power Semiconductor Devices
Formula sheet - Questions 3 & 4

$$R_{ON,SP} = \frac{W_D}{n_e q \mu_e} \tag{1}$$

$$\tau_{no} = \frac{1}{C_n N_r} = \frac{1}{v_{th} \sigma_{cn} N_r} \tag{2a}$$

and

$$\tau_{no} = \frac{1}{C_n N_r} = \frac{1}{v_{th} \sigma_{cn} N_r} \tag{2b}$$

$$n_i = \sqrt{N_C N_V} \exp\left(-\frac{E_G}{2k_B T}\right) \tag{3}$$

$$W_D = \sqrt{\frac{2\varepsilon_s V_A}{q N_D}} \tag{4}$$

$$E_{MAX} = \sqrt{\frac{2qN_D V_A}{\varepsilon_s}} \tag{5}$$

$$V_{PT} = \frac{1}{2}(E_C + E_1) t_{epi} \tag{6}$$

$$V_{PT} = E_C t_{epi} - \frac{qN_D t_{epi}^2}{2\varepsilon_s} \tag{7}$$

$$x \approx \frac{C_0 \kappa}{C_1}(t+\tau) \approx \frac{B}{A}(t+\tau) \tag{8}$$

$$x \approx \sqrt{\frac{2DC_0}{C_1}(t+\tau)} \rightarrow x^2 \approx B(t+\tau) \tag{9}$$

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} \tag{10}$$

$$C(x,t) = C_s \text{erfc}\left(\frac{x}{2\sqrt{Dt}}\right) \tag{11}$$

$$n(x) = \frac{S}{\sqrt{2\pi}\sigma_p} \exp\left(0 - \frac{(x-R_p)^2}{2\sigma_p^2}\right) \tag{12}$$

$$R_{ON,SP} = \frac{4V_B^2}{\varepsilon_s \mu_e E_{CRIT}^3} \tag{13}$$

$$T_{j(max)} = T_C + I_{ON}^2 R_{ON} R_{jc} \tag{14}$$

$$\frac{n_p}{n_{p0}} = \exp\left(\frac{q}{k_B T} V_A\right) \tag{15a}$$

and

$$\frac{p_n}{p_{n0}} = \exp\left(\frac{q}{k_B T} V_A\right) \tag{15b}$$

$$I = I_n + I_p = Aq\left(\frac{D_n n_{p0}}{l_n} + \frac{D_p p_{n0}}{l_p}\right)\left(\exp\left(\frac{qV_A}{k_B T}\right) - 1\right) \tag{16}$$

$$I = I_o\left(\exp\left(\frac{qV_A}{k_B T}\right) - 1\right) \tag{17}$$

$$I = I_o\left(\exp\left(\frac{qV_A}{nk_B T}\right) - 1\right) \tag{18}$$

$$J_n = \left(\mathscr{A}^* T^2 \exp\left(-\frac{q\phi_{Bn}}{k_B T}\right)\right) \cdot \left(\exp\left(\frac{qV_A}{k_B T}\right) - 1\right) \tag{19}$$

$$n = 1 + \left(\frac{\delta}{\varepsilon_i}\right)\frac{(\varepsilon_s / W_D) + qD_{it}}{1 + (\delta/\varepsilon_i) qD_{it}} \tag{20}$$

$$q\phi_{ms} = (q\phi_m - q\phi_s) = q\phi_m - \left(q\chi + \frac{E_G}{2} + q\psi_B\right) = 0 \tag{21}$$

$$n_p = n_i \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{22}$$

$$\psi_s(inv) = 2\psi_B = \frac{2k_B T}{q} \ln\left(\frac{N_A}{n_i}\right) \tag{23}$$

$$Q_{SC} = -qN_A W_D|_{max} = -\sqrt{2q\varepsilon_s N_A (2\psi_B)} \tag{24}$$

$$V_o = E_o d = \frac{|Q_s| d}{\varepsilon_{ox}} = \frac{|Q_s|}{C_o} \tag{25}$$

$$\frac{C}{C_o} = \frac{1}{\sqrt{1 + \frac{2\varepsilon_{ox}^2 V}{qN_A \varepsilon_s d^2}}} \tag{26}$$

continued

$$V_T = \frac{qN_A W_D}{C_o} + \psi_s\,(inv) = \frac{\sqrt{2\varepsilon_s qN_A\,(2\psi_B)}}{C_O} + 2\psi_B \quad (27)$$

$$V_{FB} = -\frac{Q_o}{C_o}\frac{d}{d} = -\frac{Q_o}{C_o} \quad (28)$$

$$I_{SD} = \beta\left[(V_{GS} - V_T)\,V_{DS} - \frac{V_{DS}^2}{2}\right] \quad (29)$$

$$I_{SD} = \frac{\beta}{2}\,(V_{GS} - V_T)^2 \quad (30)$$

$$R_{CH} = \frac{L_{CH}\,(W_m + W_t)}{2\mu C_{OX}\,(V_{GS} - V_T)} \quad (31)$$

$$R_{drift} = \frac{4V_B^2}{\mu_n \varepsilon_s E_{sMAX}^3} \quad (32)$$

$$P = \frac{fV_{DS}I_{ON}\tau}{6} \quad (33)$$

$$t_s = \frac{J_{PC}}{J_{BR}}\tau_D - \frac{1}{4}\frac{J_C}{J_{BR}}\frac{W_D^2}{qD_n} \quad (34)$$

continued

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Data sheet - Questions 3 & 4

Quantity	Symbol	Value
Boltzmann Constant	k_B	1.38066×10^{-23} J/K
Elementary Charge	q	1.60218×10^{-19} C
Electron Rest Mass	m_0	9.1095×10^{-31} kg
Electron Volt	eV	1 eV $= 1.60218 \times 10^{-19}$ J
Permittivity in Vacuum	ε_0	8.854×10^{-14} F/cm
Planck's Constant	h	6.62617×10^{-34} J s
Reduced Planck's Constant	$\hbar$	1.05458×10^{-34} J s
Proton rest mass	m_p	1.67264×10^{-27} kg
Speed of Light in Vacuum	c	2.99792×10^{10} cm/s
Thermal Voltage at 300 K	$k_B T/q$	0.0259 V
Richardson's Constant	$\mathscr{A}$	120 A cm^{-2}K^{-2}
Wavelength of 1 eV Quantum	λ	1.23977 μm

continued

Properties of Semiconductor Materials at 300 K

Properties	Si	Ge	GaAs	SiC	Diamond	SiO_2
Dielectric Constant	11.9	16	13.1	9.66	5.5	3.9
Electron Affinity (eV)	4.05	4.01	4.07	3.1-3.8	1.3	0.9
Effective Density of States in Conduction Band (cm^{-3})	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}	1.7×10^{19}	1×10^{20}	
Effective Density of States in Valence Band (cm^{-3})	2.65×10^{19}	6.0×10^{19}	7.0×10^{18}	2.5×10^{19}	1×10^{19}	
Effective Mass (Electrons) (m_0)	$m_l = 0.98$ $m_t = 0.19$	$m_l = 1.64$ $m_t = 0.082$	$m_l = 0.063$	$m_l = 0.29$ $m_t = 0.42$	$m_l = 1.40$ $m_t = 0.36$	
Effective Mass (Holes) (m_0)	$m_l = 0.16$ $m_t = 0.49$	$m_l = 0.044$ $m_t = 0.28$	$m_l = 0.076$ $m_t = 0.50$	$m_l = 1.00$ $m_t = 1.00$	$m_l = 0.80$ $m_t = 0.80$	
Energy Gap at 300 K (eV)	1.12	0.66	1.424	3.23	5.45	9.0
Critical Electric Field (MV/cm)	0.3		0.4	2.5	10	10
Intrinsic Carrier Conc. (cm^{-3})	1.45×10^{10}	2.4×10^{13}	1.79×10^{6}	10^{-9}	10^{-27}	
Minority Carrier Lifetime (s)	2.5×10^{-3}	10^{-3}	10^{-8}	6×10^{-7}		
Electron Drift Mobility ($cm^2V^{-1}s^{-1}$)	1500	3900	8500	900	1800	
Hole Drift Mobility ($cm^2V^{-1}s^{-1}$)	450	1900	400	400	1500	
Negative Differential Mob. ($cm^2V^{-1}s^{-1}$)			-100			

END